

PDF Tool

Technical Documentation — 2nd Edition

Privacy-first online PDF editor

Live: <https://pdftool4u.duckdns.org> · Guide: /help

Repository: github.com/mukeshroyhub/pdf_tool

Edition: 17 July 2026 (supersedes 16 July edition)

Owner: Mukesh Kumar · Readiness: 96/100 (Audit #6)

1. Overview

PDF Tool is a free online PDF editor with a defining privacy model: **files are stored only in the user's browser** (IndexedDB). Server tools process bytes just-in-time — staged, processed, returned, deleted in one request cycle — and the server records nothing about any user's files (test-enforced). It runs on a single stateless AWS EC2 instance behind Caddy auto-HTTPS with an Elastic IP; accounts live in Neon PostgreSQL, transient staging in Supabase S3, email via Brevo, sign-in via Google OAuth. UptimeRobot watches `/api/health`.

2. Architecture & Stack

Layer	Technology
Monorepo	pnpm workspaces (apps/web, apps/api, packages/shared). Platform-complete pnpm-lock.yaml; every environment installs <code>--frozen-lockfile</code> (reproducible builds).
Front end	Next.js 15, React 19, TypeScript, TanStack Query, Radix UI, Tailwind, pdfjs-dist viewer, Inter via next/font
API	Express 4 + TypeScript (tsx), Zod validation, JWT + rotating refresh tokens, Helmet CSP, rate limits, Multer staging
PDF engine	pdf-lib + @pdf-lib/fontkit (subset embedding), pdfjs-dist, @napi-rs/canvas, sharp, qpdf (encryption), archiver (ZIP)
Privacy layer	Browser: local-store.ts (IndexedDB files + private activity log), local-ops.ts (stage→run→absorb→cleanup with finally-guaranteed deletion). Server: 60-min staging purge, 24-h guest purge, no-op activity logger.
Fonts	3-tier matching: drop-in licensed fonts (exact match) → Inter family incl. Display optical sizes (closest match) → standard 14. Weight-granular; pixel-sampled color; real font-data bold/italic detection.
Infra	Docker Compose (api/web/caddy, non-root), Caddy 2 auto-TLS, DuckDNS + Elastic IP 13.50.135.60, t3.micro + 3 GB swap + 30 GB gp3
CI (blocking)	typecheck · API unit tests · web build · 4-journey Playwright E2E (guest/upload/viewer, page numbers, protect+unlock, merge) — pnpm frozen installs

3. Drop-in Licensed Fonts (operations)

For exact-match text editing, licensed font files live on the HOST (never in git, never served to browsers): `~/pdf_tool/fonts-custom/`, volume-mounted read-only into the API container. Missing files degrade gracefully to Inter/Helvetica.

- **Format requirement:** use TTF. CFF-flavored OTF subsets are rejected by pdfium (Edge/Chrome) and poppler, which substitute a wrong font — verified across renderers, July 2026. Convert OTFs with fontTools' `otf2ttf` (cu2qu) before installing.
- **Install:** `scp` the .tff files into `fonts-custom/` on the box, then `docker compose -f docker-compose.oracle.yml restart api` (clears the font cache; no rebuild needed).
- **Mapping:** `apps/api/src/lib/fonts.ts` `DROPIN_FILES` — keys (e.g. `ubermove`, `ubermove-medium`, `ubermove-bold`) list candidate filenames in preference order; the client matcher routes detected font names/weights to keys.

- **Robustness:** the server drops characters the chosen font cannot draw (prevents invisible .notdef 'phantom spaces'); the editor repairs common extraction artifacts (spaces before punctuation, digit:digit separators, garbled non-ASCII between digits) on pre-fill.

4. Deployment & Operations

4.1 Deploy / update

```
cd ~/pdf_tool && git pull && docker compose -f docker-compose.oracle.yml up -d --build
```

10–20 min on 1 GB (swap carries the Next.js build). Config-only change: `up -d --force-recreate`. Disk hygiene before big builds: `docker system prune -af` (keep ≥ 5 GB free).

4.2 Local development (Windows)

```
corepack enable && pnpm install && pnpm dev # web :3000 + api :4000
```

After any node_modules wipe: `pnpm --filter @pdfforge/api db:generate`. E2E locally: `pnpm exec playwright install chromium && pnpm test:e2e`.

4.3 Scheduled jobs & monitoring

- Token purge 12 h · staged-data purge 15 min (60-min retention) · stale-guest purge 24 h.
- UptimeRobot: `/api/health` every 5 min with alerts. Health sits above the rate limiter.

4.4 Known limitations

- Single instance (recovery = rebuild; see DR-RUNBOOK.md). EC2 free tier ends mid-2027 (~\$8–10/mo after).
- Browser-local files are per-device; clearing site data deletes them (documented in UI).
- Poppler-based viewers (Evince) may substitute fonts on edited text; spacing preserved via `/Widths`.
- Save round-trip 2–5 s — inherent to just-in-time privacy processing.

5. Release History (July 2026)

Release	Contents
Security & services	6 audit fixes; Neon/Supabase/Brevo/Google; proxy-safe OAuth
Features & UX	Page numbers, ZIP, protect/unlock, e-signatures; dark mode, tool grid, auto-open
EC2 migration	Render → EC2, Caddy HTTPS, DuckDNS; swap/disk; UptimeRobot; later Elastic IP
Privacy architecture	Browser-local library + stateless processing; server activity logging removed (test-enforced)
UI & viewer polish	Upload-first dashboard, Tools dropdown, password prompt, image preview, resize handles
Editor fidelity	Pixel-sampled color, real-font-data weights, Inter family + Display, serif/oblique variants
Ops hardening	Non-root containers, sitemap/robots, privacy card, Playwright gate, secrets rotated, Render retired
Reproducible builds	pnpm migration: platform-complete lockfile, frozen installs, phantom-dependency fix

Drop-in fonts	Licensed font layer (Uber Move), TTF conversion pipeline, cross-renderer verification, unglyphed-char sanitization
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